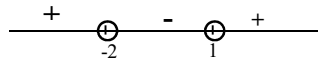
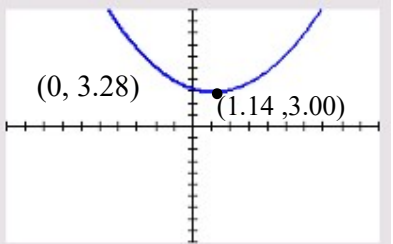
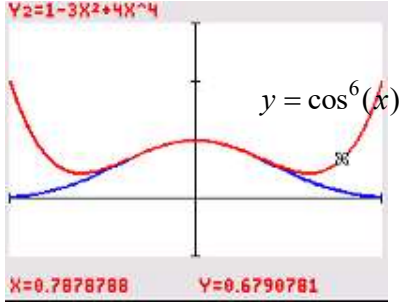


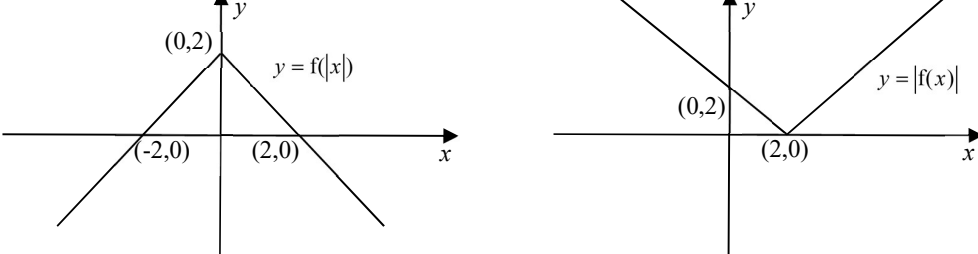
2011 'A' Level H2 Mathematics Paper 1 9740/01

Qn 1	Suggested Solutions	Do a quick check if the 2 quadratic expressions can be factorised into linear factors. If not, either complete the square or use the discriminant method to explain that it is always positive or always negative. ** Take note of presentation
Must justify that the number is positive for all real values of x.	$x^2 + x + 1 = (x + \frac{1}{2})^2 + \frac{3}{4} > 0$, for all $x \in \mathbb{R}$ <u>Alternative</u> Discriminant of $x^2 + x + 1$ $= 1^2 - 4(1)(1) = -3$ Since coefficient of $x^2 > 0$, $x^2 + x + 1 > 0$ for all $x \in \mathbb{R}$ $\frac{x^2 + x + 1}{x^2 + x - 2} < 0$ $\frac{1}{x^2 + x - 2} < 0$, $\therefore x^2 + x + 1 > 0$ for all $x \in \mathbb{R}$ $\frac{1}{(x+2)(x-1)} < 0$, $x \neq -2, x \neq 1$ $\therefore -2 < x < 1$ 	

Qn 2	Suggested Solutions	**Presentation + Read Qn Carefully Give answers to the precision required by the question. In this case, its correct to 3 decimal places.
(i)	Since $y = f(x)$ passes through $(-1.5, 4.5)$, $(2.1, 3.2)$ and $(3.4, 4.1)$, we can form 3 equations, $4.5 = a(-1.5)^2 - 1.5b + c$ (1) $3.2 = a(2.1)^2 + 2.1b + c$ (2) $4.1 = a(3.4)^2 + 3.4b + c$ (3) Solving using GC, we have $a = 0.215$, $b = -0.490$, $c = 3.281$ (to 3 d.p.)	
(ii)	 For $f(x)$ to be an increasing function, we can either use differentiation to show $f'(x) > 0$ or use GC to sketch the curve and find the minimum point. Since it is a 2 marks question, the faster way will be to use GC. From the graph, the set of values for which $f(x)$ is an increasing function is $\{x \in \mathbb{R} : x > 1.14\}$.	

Qn 3	Suggested Solutions	
(i)	$x = t^2, y = \frac{2}{t}$ $\frac{dx}{dt} = 2t, \frac{dy}{dt} = -\frac{2}{t^2}$ $\therefore \frac{dy}{dx} = \left(-\frac{2}{t^2}\right) \cdot \left(\frac{1}{2t}\right) = -\frac{1}{t^3}$ At the point $\left(p^2, \frac{2}{p}\right)$, $t = p$. $\therefore \frac{dy}{dx} = -\frac{1}{p^3}$ Equation of tangent at $\left(p^2, \frac{2}{p}\right)$ is $y - \frac{2}{p} = -\frac{1}{p^3}(x - p^2)$ $p^3 y + x = 3p^2.$	
(ii)	At point Q, $y = 0$. $\Rightarrow x = 3p^2.$ At point R, $x = 0$. $\Rightarrow y = \frac{3}{p}.$ $\therefore$ The coordinates are $Q(3p^2, 0)$ and $R\left(0, \frac{3}{p}\right)$.	
(iii)	Midpoint of QR is $\left(\frac{3p^2}{2}, \frac{3}{2p}\right)$. $x = \frac{3p^2}{2} \text{ ----- (1)}$ $y = \frac{3}{2p} \Rightarrow p = \frac{3}{2y} \text{ ----- (2)}$ Sub (2) into (1): $x = \frac{3}{2} \left(\frac{3}{2y}\right)^2 = \frac{27}{8y^2}$ $\therefore$ the Cartesian equation is $8xy^2 = 27$.	As p varies, we get different points on the curve. These points when joined together will form a curve given by the Cartesian equation.

Qn 4	Suggested Solutions
(i)	From MF26, the Maclaurin series for $\cos x$ is $1 - \frac{x^2}{2} + \frac{x^4}{24} + \dots$ $g(x) = \cos^6 x$ $= \left(1 - \frac{x^2}{2} + \frac{x^4}{24} + \dots\right)^6$ $= 1 + 6\left(-\frac{x^2}{2} + \frac{x^4}{24} + \dots\right) + 15\left(-\frac{x^2}{2} + \frac{x^4}{24} + \dots\right)^2 + \dots$ $= 1 - 3x^2 + \frac{1}{4}x^4 + 15\left(\frac{1}{4}x^4\right) + \dots$ $= 1 - 3x^2 + 4x^4 + \dots$ Make use of the formula in MF26. To find $g(x)$, group $-\frac{x^2}{2} + \frac{x^4}{24}$ as a term and carry out binomial expansion.
(ii) (a)	$\int_0^a g(x) dx \approx \int_0^a (1 - 3x^2 + 4x^4) dx = \left[x - x^3 + \frac{4}{5}x^5\right]_0^a$ $= a - a^3 + \frac{4}{5}a^5$ Substitute $a = \frac{1}{4}\pi$, $\int_0^{\frac{1}{4}\pi} g(x) dx \approx \frac{1}{4}\pi - \left(\frac{1}{4}\pi\right)^3 + \frac{4}{5}\left(\frac{1}{4}\pi\right)^5 = 0.540$
(ii) (b)	From GC, $\int_0^{\frac{1}{4}\pi} g(x) dx = 0.475$ (to 3 s.f.) The value of $\frac{1}{4}\pi$ is approximately 0.785, and at $x = 0.785$, the graphs of $y = 1 - 3x^2 + 4x^4$ and $y = g(x)$ are significantly different, as seen in the graph below. The graph of $y = 1 - 3x^2 + 4x^4$ diverges from the graph of $g(x) = \cos^6 x$ at $x = \frac{1}{4}\pi$. Hence the approximation in part (ii)(a) is not very good.  Note that the Maclaurin series is a good approximation for a function in the neighbourhood of 0. Beyond a certain interval, the approximation will not be a good one (as shown in the diagram). To improve the approximation, we can generate more terms of the series.

Qn 5	Suggested Solutions
(i)	
(ii)	$\{x \in \mathbb{R} : 0 \leq x \leq 2\}$
(iii)	Using the graph in part (i), $\int_{-1}^1 f(x) \, dx = 3$ Since $\int_1^2 f(x) \, dx = \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$, $a > 2$ $\int_1^a f(x) \, dx = \frac{1}{2} + \frac{1}{2}(a-2)(a-2) = 3$ $\Rightarrow \frac{1}{2} + \frac{1}{2}(a-2)^2 = 3$ $\Rightarrow a = 2 \pm \sqrt{5}$ Since $a > 2$, $a = 2 + \sqrt{5}$ Always try and relate different parts of a question. In this part, make use of the graphs drawn in (i) and use areas of triangles and trapezia to solve.

Qn 6	Suggested Solutions
(i)	$\sin\left(r + \frac{1}{2}\right)\theta - \sin\left(r - \frac{1}{2}\right)\theta$ $= \sin r \theta \cos \frac{1}{2}\theta + \cos r \theta \sin \frac{1}{2}\theta - \left(\sin r \theta \cos \frac{1}{2}\theta - \cos r \theta \sin \frac{1}{2}\theta\right)$ $= 2 \cos r \theta \sin \frac{1}{2}\theta \quad (\text{proved})$ **Make use of the formula in MF26.

(ii)	$\sum_{r=1}^n \cos r\theta = \frac{1}{2 \sin \frac{1}{2}\theta} \sum_{r=1}^n \left(\sin \left(r + \frac{1}{2} \right) \theta - \sin \left(r - \frac{1}{2} \right) \theta \right)$ $= \frac{1}{2 \sin \frac{1}{2}\theta} \left(\sin \frac{3}{2}\theta - \sin \frac{1}{2}\theta \right.$ $\quad + \sin \frac{5}{2}\theta - \sin \frac{3}{2}\theta$ $\quad + \sin \frac{7}{2}\theta - \sin \frac{5}{2}\theta$ $\quad \dots$ $\quad + \sin \left(n - \frac{1}{2} \right) \theta - \sin \left(n - \frac{3}{2} \right) \theta$ $\quad \left. + \sin \left(n + \frac{1}{2} \right) \theta - \sin \left(n - \frac{1}{2} \right) \theta \right)$ $= \frac{1}{2 \sin \frac{1}{2}\theta} \left(\sin \left(n + \frac{1}{2} \right) \theta - \sin \frac{1}{2}\theta \right)$
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Qn 7	Suggested Solutions
(i)	$\overrightarrow{OP} = \frac{1}{3}\mathbf{a}, \quad \overrightarrow{OQ} = \frac{3}{5}\mathbf{b}$ $\overrightarrow{OM} = \frac{\overrightarrow{OP} + \overrightarrow{OQ}}{2} = \frac{1}{6}\mathbf{a} + \frac{3}{10}\mathbf{b}$ $\text{Area} = \frac{1}{2} \overrightarrow{OP} \times \overrightarrow{OM} $ $= \frac{1}{2} \left \frac{1}{3}\mathbf{a} \times \left(\frac{1}{6}\mathbf{a} + \frac{3}{10}\mathbf{b} \right) \right $ $= \frac{1}{2} \left \frac{1}{3}\mathbf{a} \times \frac{1}{6}\mathbf{a} + \frac{1}{3}\mathbf{a} \times \frac{3}{10}\mathbf{b} \right $ $= \frac{1}{20} \mathbf{a} \times \mathbf{b} $ $\therefore k = \frac{1}{20}$ Note: $\mathbf{a} \times \mathbf{a} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ Alternative Method: Area of OMP $= \frac{1}{2} \times \text{Area of OPQ}$ $= \frac{1}{2} \times \frac{1}{2} \left \frac{1}{3}\mathbf{a} \times \frac{3}{5}\mathbf{b} \right$ $= \frac{1}{20} \mathbf{a} \times \mathbf{b}$
(ii)(a)	$ \mathbf{a} = 1$ $\sqrt{(2p)^2 + (-6p)^2 + (3p)^2} = 1$ $49p^2 = 1$ $p = \frac{1}{7} \quad \left(\text{reject } p = -\frac{1}{7} \text{ since } p > 0 \right)$
(b)	$ \mathbf{a} \cdot \mathbf{b} $ is the length of projection of $\mathbf{b}$ onto $\mathbf{a}$. (Note: $\mathbf{a}$ is a unit vector)

Qn 7	Suggested Solutions
(c)	$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} 2p \\ -6p \\ 3p \end{pmatrix} \times \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} = p \begin{pmatrix} 2 \\ -6 \\ 3 \end{pmatrix} \times \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} = \frac{1}{7} \begin{pmatrix} 9 \\ 7 \\ 8 \end{pmatrix}$

Qn 8	Suggested Solutions
(i)	$\int \frac{1}{100-v^2} dv = \int \frac{1}{10^2-v^2} dv = \frac{1}{20} \ln \left \frac{10+v}{10-v} \right + c$
(ii)(a)	$\frac{dv}{dt} = 10 - 0.1v^2$ $10 \frac{dv}{dt} = 100 - v^2$ $\int \frac{1}{100-v^2} dv = \frac{1}{10} \int \frac{1}{1-v^2/100} dt$ $\frac{1}{20} \ln \left \frac{10+v}{10-v} \right = \frac{1}{10} t + c$ $\ln \left \frac{10+v}{10-v} \right = 2t + 20c$ $\left \frac{10+v}{10-v} \right = e^{2t+20c}$ $\frac{10+v}{10-v} = Ae^{2t}, A = \pm e^{20c}$ When $t=0, v=0$, $\frac{10+0}{10-0} = A$, hence $A=1$ $\frac{10+v}{10-v} = e^{2t}$ $\therefore t = \frac{1}{2} \ln \left(\frac{10+v}{10-v} \right)$ When $v=5$, $t = \frac{1}{2} \ln \left(\frac{10+5}{10-5} \right) = \frac{1}{2} \ln 3$
(b)	After 1 sec ($t=1$), $e^2 = \left(\frac{10+v}{10-v} \right)$ $10e^2 - e^2v = 10+v$ $v = \frac{10e^2 - 10}{1 + e^2} = 7.6159 \approx 7.62$ The speed of the stone is 7.62m/s.

Qn 8	Suggested Solutions
(c)	Since $e^{2t} = \left(\frac{10+v}{10-v}\right)$ $10e^{2t} - e^{2t}v = 10 + v$ $v = \frac{10e^{2t} - 10}{1 + e^{2t}} = \frac{10(e^{2t} - 1)}{e^{2t} + 1} = \frac{e^{-2t}10(e^{2t} - 1)}{e^{-2t}(e^{2t} + 1)} = \frac{10(1 - e^{-2t})}{1 + e^{-2t}}$ Hence when $t \rightarrow \infty, e^{-2t} \rightarrow 0$ $\therefore v \rightarrow \frac{10(1-0)}{1+0} = 10$ The speed of the stone for large values of t tends to 10m/s. 'For large value of t' means $t \rightarrow \infty$.

Qn 9	Suggested Solutions
(i)	$a = 256, d = -7$ $T_{10} = 256 + 9(-7) = 193$ $T_n < 10$ $256 + (n-1)(-7) < 10$ $256 - 7n + 7 < 10$ $7n > 253$ $n > 36.143$ Least $n = 37$ $S_{37} = \frac{37}{2}[2(256) + 36(-7)] = 4810$ $\therefore$ The total depth is 4810m when drilling is completed.
(ii)	$a = 256, r = \frac{8}{9}$ $S_n > 0.99S_{\infty}$ $\frac{a}{1-r} \left(1 - \frac{8^n}{9}\right) > (0.99) \frac{a}{1-r}$ $\left(1 - \left(\frac{8}{9}\right)^n\right) > (0.99) \quad \therefore \frac{a}{1-r} > 0$ $\left(\frac{8}{9}\right)^n < 0.01$ $n \ln\left(\frac{8}{9}\right) < \ln 0.01$ $n > 39.099$ It will take 40 days. Note: $\ln\left(\frac{8}{9}\right) < 0$

Qn 10	Suggested Solutions
(i)	$z^2 = -8i$ Let $z = x + iy$ $(x + iy)^2 = -8i$ $x^2 - y^2 + 2xyi = -8i$ Comparing the real and imaginary parts, $x^2 - y^2 = 0 \quad 2xy = -8 \Rightarrow y = -\frac{4}{x}$ $x^2 - \left(-\frac{4}{x}\right)^2 = 0$ $x = \pm 2$ when $x = 2, y = -2$ when $x = -2, y = 2$ $\therefore z_1 = 2 - 2i \quad \text{or} \quad z_2 = -2 + 2i$ Alternative Method: $z^2 = 8e^{i\left(-\frac{\pi}{2}\right)}$ $z^2 = 8e^{i\left(-\frac{\pi}{2} + 2k\pi\right)} \quad \text{where } k = 0, 1.$ $z = 2\sqrt{2}e^{i\left(-\frac{\pi}{4} + k\pi\right)}$ $z = 2\sqrt{2}e^{i\left(-\frac{\pi}{4}\right)} \quad \text{or} \quad 2\sqrt{2}e^{i\left(\frac{3\pi}{4}\right)}$ $z_1 = 2 - 2i \quad \text{or} \quad z_2 = -2 + 2i$
(ii)	$w^2 + 4w + (4 + 2i) = 0$ $w = \frac{-4 \pm \sqrt{4^2 - 4(4 + 2i)}}{2}$ $w = -2 \pm \frac{\sqrt{-8i}}{2}$ Using result from (i): $z^2 = -8i$ $z = 2 - 2i \quad \text{or} \quad -2 + 2i$ $w = -2 + \frac{2 - 2i}{2} \quad \text{or} \quad -2 + \frac{-2 + 2i}{2}$ $w = -1 - i \quad \text{or} \quad -3 + i$ Alternative Method: $w^2 + 4w + (4 + 2i) = 0$ $(w + 2)^2 = -2i$ $(w + 2)^2 = \frac{z^2}{4}$ From (i): $z^2 = -8i$ $w + 2 = \frac{2 - 2i}{2} \quad \text{or} \quad \frac{-2 + 2i}{2}$ $w = -1 - i \quad \text{or} \quad -3 + i$
Qn 11	Suggested Solutions
(i)	Two vectors parallel to p are $\begin{pmatrix} 4 \\ -1 \\ -3 \end{pmatrix} - \begin{pmatrix} -2 \\ -5 \\ 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \\ -5 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ -1 \\ -3 \end{pmatrix} - \begin{pmatrix} 4 \\ -3 \\ -2 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}$. Thus $\begin{pmatrix} 6 \\ 4 \\ -5 \end{pmatrix} \times \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 6 \\ 6 \\ 12 \end{pmatrix}$ and a normal vector of p can be taken as $\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$.

	Thus, a scalar product form of p is given as $\mathbf{r} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = -3$. Let $\mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, and we have $\begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = -3$. Hence in Cartesian form, the equation is $x + y + 2z = -3$.
(ii)	Convert equations of both lines to vector form: $l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix}, \quad \lambda \in \mathbb{R} \quad \text{and} \quad l_2 : \mathbf{r} = \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 5 \\ k \end{pmatrix}, \quad \mu \in \mathbb{R}$ Since the two lines intersect, $\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 5 \\ k \end{pmatrix}$ $1 + 2\lambda = -2 + \mu \text{ -----(1)}$ $2 - 4\lambda = 1 + 5\mu \text{ -----(2)}$ $-3 + \lambda = 3 + k\mu \text{ -----(3)}$ Solving (1) and (2) gives $\lambda = -1$ and $\mu = 1$. Substitute these values into (3) to get $k = -7$.
(iii)	Let $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix}$ which is a vector representing any point on l_1. Denote $p : \mathbf{r} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = -3 \text{ -----(1)}$ Substitute this vector into LHS of (1): $\left[\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix} \right] \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = -3 + \lambda(0) = -3 \text{ (RHS of (1))}$ Thus l_1 lies on p.

	Since l_1 lies in p, then the intersection point of l_2 and p is the intersection point of l_2 and l_1. From (ii) Substitute $\lambda = -1$ into l_1, $\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} - (1) \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 6 \\ -4 \end{pmatrix}$ Thus, the coordinates of intersection point is $(-1, 6, -4)$. Alternatively, To find the point of intersection of l_2 and p, $\left[\begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 5 \\ -7 \end{pmatrix} \right] \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = -3$ $\begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 5 \\ -7 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = -3$ $5 - 8\mu = -3$ $\mu = 1$ Use this value on $r = \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 5 \\ -7 \end{pmatrix}$ to obtain $r = \begin{pmatrix} -1 \\ 6 \\ -4 \end{pmatrix}$.
(iv)	$\sin \theta = \frac{ \mathbf{n} \cdot \mathbf{d} }{\ \mathbf{n}\ \ \mathbf{d}\ } = \frac{\left \begin{pmatrix} 1 \\ 5 \\ -7 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \right }{\sqrt{75} \sqrt{6}} = \frac{8}{\sqrt{450}}$ $\therefore \theta = 22.2^\circ$ (1 d.p.) For angle, answer must be corrected to 1 decimal place.